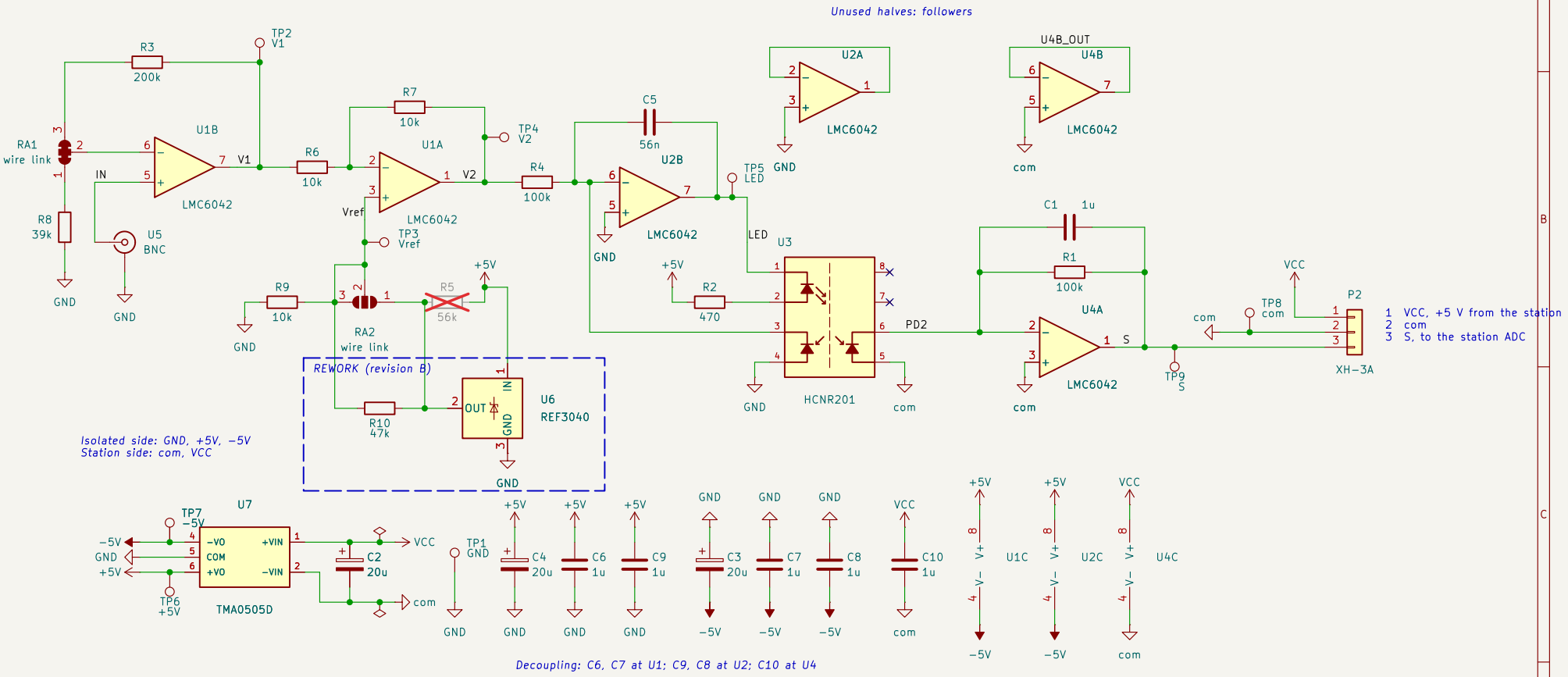
pH electrode front end, release variant (no trimmers)

1. U1 pin 5 is bent up and wired in the air straight to the BNC centre: no pad, no copper.
2. RA1, RA2: no trimmers. A wire joins all three holes of each.
3. First stage gain 1 + R3/R8 = 6.13.
4. REWORK (verified revision): R5 not fitted (DNP). REF3040AIDBZR (4.096 V) on the R5 pads: IN to +5V, GND to ground, OUT to RA2 pin 1. 47k (R10) across RA2 holes 1–2, wire RA2 2–3. Vref = 4.096 V * R9/(47k + R9) = 0.719 V (measured 0.688 V). Unreworked board: Vref = V(+5V) * R9/(R5 + R9) = 0.152 * V(+5V).
5. Design: S = 0.5 V at pH 4 ... 2.45 V at pH 9.38, slope −6.13 V/V (measured with the REF3040 rework: −6.09 V/V).
6. C1 and C5: ceramic only. C2–C4: 4.7–47 uF, 10 V or more.



Rework (U6, R10): igorlab
Hardware licence: CERN−OHL−S−2.0
Release variant (no trimmers); values as fitted on the working board.
igorlab

Sheet: /
File: ph_board.kicad_sch

Title: pH electrode front end: LMC6042 + HCNR201 + TMA0505D

Size: A4	Date: 2026−10−21	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 1/1